

Abstract

Urban transportation management systems (UTMS) depend on sensor and event data that are often incomplete, duplicated, or schema-inconsistent before any forecasting or anomaly workflow can run. This thesis designs, implements, and evaluates **Flowmatic**: an intelligent preparation and inference platform that automates quality assessment, routes prepared streams to modality-specific neural checkpoints, and publishes reproducible training artifacts.

Flowmatic implements a six-dimensional Data Quality Index (DQI), automated cleaning, and export in a containerised web stack. In batch platform validation, a 30 000-row Astana upload completes in 406 ms; under controlled corruptions up to 20%, composite DQI recovers by up to 0.045 while invalid rows are removed. A smart-city workbench connects simulated traffic and weather sources, Manual/Auto routing over a documented registry, and medallion lake export.

Seven primary neural checkpoints are trained under a shared temporal hold-out protocol (70/15/15 split, three seeds). The severity classifier reaches macro-F1 = 0.911 ± 0.017 , exceeding sklearn baselines on the same held-out test partition; labels are rule-based synthetic attributes. TranAD achieves AUROC = 0.84 on injected window anomalies. A preparation ablation lowers PatchTST density RMSE from 0.90 to 0.58 after 20% corruption when corrupted cells are repaired. Rule-based Auto routing over 140 simulator events achieves 100% scenario-aligned policy conformance under documented oracles, not independent expert-labelled routing accuracy.

Artifacts are published on Hugging Face in the `pushthetempo/flowmatic-*` repository family. Batch API validation uses the Astana export only; live agency deployment and external routing labels remain future work.

Keywords: Data Preparation; Intelligent Transportation Systems; Smart Cities; Flowmatic; Model Routing; Time Series; Reproducibility